

Session 12: Post Class tests

- Gerard Enterprises is a publicly traded company. You are trying to estimate how much debt it has outstanding, to compute a cost of capital. Which of the following items would you not include in debt and why?
 - Short term bank loans
 - Commercial paper
 - Corporate bonds
 - Deferred Tax Liabilities
 - None of the above
- Lipscott Inc. is a publicly traded company that has \$100 million in bank loan on its books, with a stated interest rate of 3% and \$150 million in publicly traded bonds, with a coupon rate of 3.6%. The company currently has a bond rating of BBB, with a default spread of 1.5% over the risk free rate. If the current T.Bill rate is 1%, the ten-year T.Bond rate is 3.5% and the marginal tax rate is 35%, what is the pre-tax cost of debt?
 - 3.36%
 - 3.60%
 - 5.00%
 - 2.50%
 - 3.50%
- Ricotta Inc. is a food processing company that generated \$80 million in operating income in the most recent year. The company has \$300 million in bank loans with an average book interest rate of 15%. The company is not rated but you have the following table to compute a bond rating:

Interest coverage ratios and Default Spreads

<i>Large cap (>\$5 billion)</i>	<i>Small cap or risky (<\$5 billion)</i>	<i>Rating is (S&P/Moody's)</i>	<i>Default Spread</i>
>8.50	>12.5	Aaa/AAA	0.40%
6.5-8.5	9.5-12.5	Aa2/AA	0.70%
5.5-6.5	7.5-9.5	A1/A+	0.85%
4.25-5.5	6-7.5	A2/A	1.00%
3-4.25	4.5-6	A3/A-	1.30%
2.5-3	4-4.5	Baa2/BBB	2.00%
2.25-2.5	3.5-4	Ba1/BB+	3.00%
2-2.25	3-3.5	Ba2/BB	4.00%
1.75-2.25	2.5-3	B1/B+	5.50%
1.5-1.75	2-2.5	B2/B	6.50%
1.25-1.5	1.5-2	B3/B-	7.25%
0.8-1.25	1.25-1.5	Caa/CCC	8.75%
0.65-0.8	0.8-1.25	Ca2/CC	9.50%
0.2-0.65	0.5-0.8	C2/C	10.50%
<0.2	<0.5	D2/D	12.00%

If the risk free rate is 3% and Ricotta Inc. has 100 million shares trading at \$15/share, estimate the pre-tax cost of debt for the company.

- a. 4.00%
 - b. 4.30%
 - c. 5.00%
 - d. 6.30%
 - e. None of the above
4. KGV Inc. has \$ 1 billion in bonds outstanding, with a coupon rate of 4%. The company is rated BBB, with a default spread of 2% (on top of a risk free rate of 3%). The company has an effective tax rate of 20% and the marginal tax rate is 40%. What is the after-tax cost of debt for the company?
- a. 2.4%
 - b. 3.0%
 - c. 3.2%
 - d. 4.0%
 - e. 5.0%
5. Garana Inc. is a Colombia coffee company. You have computed a synthetic rating of BBB (with a default spread of 2%) for the company. The US treasury bond rate is 3%, the Colombian Government US \$ bond rate is 4.5% and the Colombian Government Peso bond rate is 6%. What is the US\$ pre-tax cost of debt for Garana?
- a. 3.00%
 - b. 4.5%
 - c. 5.0%
 - d. 6.5%
 - e. 8.0%

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1. **d. Deferred tax liabilities.** These are not legal commitments in the conventional sense but accounting liabilities (reflecting expectations that the firm will have to pay more in taxes in the future).
2. **d. 5.00%.**
The pre-tax cost of debt is a long term cost of borrowing money today.
 - Pre-tax cost of debt = $3.5\% + 1.5\% = 5\%$
3. **c. 4.30%.**
Interest coverage ratio = $80/15 = 5.33$
Market cap = \$1.5 billion
Synthetic rating based on rating = A-
Default spread for A- rating = 1.30%
Pre-tax cost of debt = $3\% + 1.30\% = 4.30\%$
4. **b. 3.0%**
Pre-tax cost of debt = Riskfree rate + Default spread = $3\% + 2\% = 5\%$
After-tax cost of debt = $5\% (1-.40) = 3\%$
5. **d. 6.5%**
Risk free rate in US \$ = 3%
Default spread for Colombia = $4.5\% - 3\% = 1.5\%$
Default spread for company = 2%
Cost of debt in US \$ = $3\% + 1.5\% + 2\% = 6.5\%$